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Mathematics Extension 1

HSC ASSESSMENT TASK 3

Week 6 2014

Weighting of task 20%

General Instructions

- Reading time – 5 minutes
- Working time – 1 hour
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In questions 6–7, show relevant mathematical reasoning and/or calculations

Total marks – 35

Section I Pages 2 - 3

5 marks

- Attempt Questions 1 – 5
- Allow about 8 minutes for this section

Section II Pages 4 - 7

30 marks

- Attempt Questions 6 – 7
- Allow about 52 minutes for this section

THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

		MARKS
Questions 1 – 5	Objective Response	/5
Question 6	Inverse Functions	/18
Question 7	Mathematical Induction	/3
	Permutations and Combinations	/9
Total		/35

Section I

5 marks

Attempt Questions 1-5

Allow about 8 minutes for this section

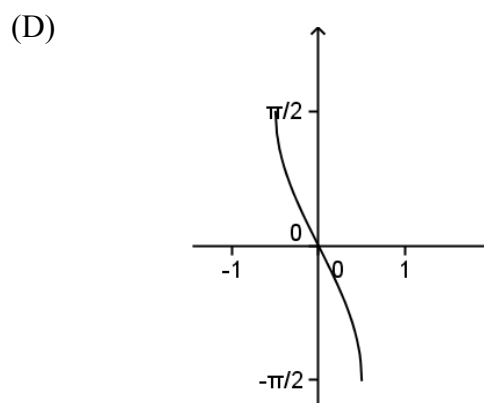
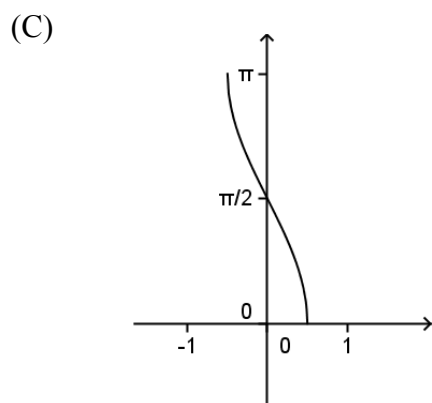
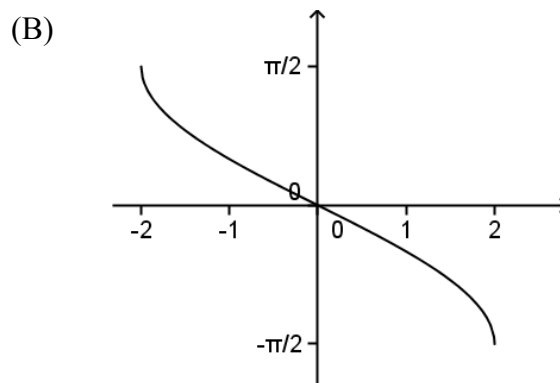
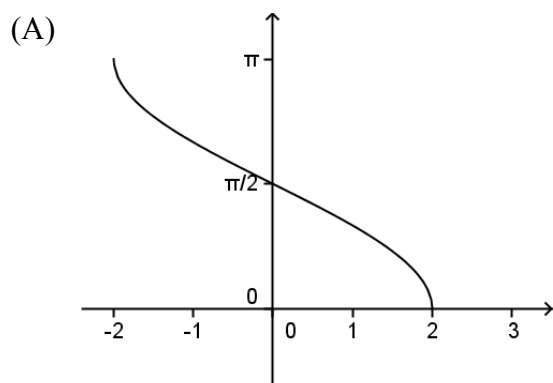
Use the multiple-choice answer sheet for Questions 1-5

- 1 The point C divides the interval joining A $(2j, -k)$ and B $(j, 3k)$ internally in the ratio 3 : 5. 1

What are the coordinates of C?

- (A) $(\frac{11j}{8}, \frac{3k}{2})$
(B) $(\frac{13j}{8}, \frac{k}{2})$
(C) $(\frac{-7j}{8}, \frac{7k}{4})$
(D) $(\frac{j}{8}, \frac{-9k}{4})$

- 2 Which diagram shows the graph of $y = \cos^{-1} \frac{x}{2}$? 1



3 Evaluate the $\lim_{x \rightarrow 0} \frac{\sin 4x}{10x}$ **1**

- (A) 0
- (B) 0.4
- (C) 1
- (D) 4

4 For all integers $n \geq 1$, $2^{3n} - 7n - 1$ is divisible by **1**

- (A) 0
- (B) 49
- (C) 98
- (D) all of the above

5 Which function does NOT have the same graph as $y = \cos^{-1}(\cos x)$ in the domain $-\pi < x < \pi$? **1**

- (A) $y = \sin^{-1}(\sin(x - \frac{\pi}{2})) + \frac{\pi}{2}$
- (B) $y = |\tan(\tan^{-1} x)|$
- (C) $y = |x|$
- (D) $y = \frac{\pi}{2}(1 - \cos x)$

Section II

30 marks

Attempt Questions 6 – 7

Allow about 52 minutes for this section

Answer each question on a SEPARATE sheet of paper.
All necessary working should be shown in every question.

Question 6 (18 marks) Use a SEPARATE sheet of paper.

Marks

- (a) (i) Sketch the graph of the function $f(x) = \frac{3x^2}{x^2-12}$ showing any intercepts and asymptotes. **3**
- (ii) Explain why $f(x)$ does not have an inverse function. **1**
- (iii) Show that equation of the inverse relation of $f(x)$ can be written as $y = \pm \sqrt{\frac{12x}{x-3}}$ **1**
- (iv) State the domain of the inverse relation $y = \pm \sqrt{\frac{12x}{x-3}}$ **1**
- (b) Find $\frac{d}{dx}(x \cos^{-1} 2x)$ **2**
- (c) Find the volume of the solid formed when the region between $y = (1 - 25x^2)^{-\frac{1}{4}}$ and the x-axis from $x = -\frac{1}{10}$ to $x = \frac{1}{10}\sqrt{3}$ is rotated about the x-axis. **3**

Question 6 continues

YR12 X1 Task 3 -1- Solutions

Start here Multiple Choice

1. $\frac{8!}{3!2!} = 3360$ (B)

2. $\int_0^3 (1-x)^{-2} dx = -\int_0^3 (1-x)^{-2} (-dx)$ (A)

3. $(2, -\frac{1}{4})$ (C)

$= \left[-\frac{1}{1-x} \right]_0^3 = \left[\frac{1}{1-3} - 1 \right] = -\frac{3}{2}$

4. $\cos x = \frac{\sqrt{3}}{2}$ (B)

5. $\frac{1 - \frac{1-t^2}{1+t^2}}{\frac{2t}{1+t^2}} = \frac{1-t^2-1+t^2}{2t} = t$

$x = 2n\pi \mp \frac{\pi}{6}$

(B)

Section 2

6. a) $\sin^{-1}(\sin \frac{4\pi}{3}) = \sin^{-1}(-\frac{\sqrt{3}}{2})$

2 marks for correct solution

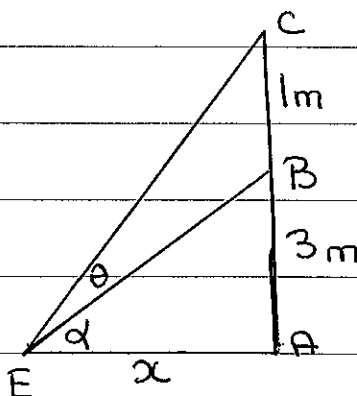
$= -\frac{\pi}{3}$

1 mark substitutes $\sin \frac{4\pi}{3} = -\frac{\sqrt{3}}{2}$

b i) $\triangle ACE$

$\tan(\theta + d) = \frac{4}{x}$

$\theta + d = \tan^{-1} \frac{4}{x}$



2 marks correct working out

$\triangle ABE$ $\tan d = \frac{3}{x}$

$d = \tan^{-1} \frac{3}{x}$

1 mark shows $\tan d$ or $\tan(\theta + d)$

$\therefore (\theta + d) - (d) = \theta = \tan^{-1} \frac{4}{x} - \tan^{-1} \frac{3}{x}$

ii)

Question 6 continues...

ii) At maximum point $\frac{d\theta}{dx} = 0$

Using $\frac{d}{dx} \tan^{-1} u = \frac{1}{1+u^2} \frac{du}{dx}$

$$\frac{d\theta}{dx} = \frac{1}{1+\frac{16}{x^2}} \times \left(\frac{-4}{x^2}\right) - \frac{1}{1+\frac{9}{x^2}} \times \left(\frac{-3}{x^2}\right)$$

$$\frac{d\theta}{dx} = \frac{-4}{x^2+16} + \frac{3}{x^2+9}$$

$$= \frac{-4x^2-36+3x^2+48}{(x^2+16)(x^2+9)}$$

$$\frac{d\theta}{dx} = \frac{-x^2+12}{(x^2+16)(x^2+9)} \quad \text{at stationary pt } \frac{d\theta}{dx} = 0$$

$$x^2 = 12, \quad x = 2\sqrt{3} > 0$$

The nature of turning pt:

$$\frac{d^2\theta}{dx^2} = \frac{-4(-2x)}{(x^2+16)^2} + \frac{-3(2x)}{(x^2+9)^2} = \frac{8x}{(x^2+16)^2} - \frac{6x}{(x^2+9)^2}$$

When $x = 2\sqrt{3}$ $\frac{d^2\theta}{dx^2} = \frac{8 \times 2\sqrt{3}}{(12+16)^2} - \frac{6 \times 2\sqrt{3}}{(12+9)^2}$

$$\frac{d^2\theta}{dx^2} = -0.01178 \quad \therefore \theta \text{ is maximum when } x = 2\sqrt{3}$$

2 marks correct answer

1 mark differentiates θ with respect to x or any attempt towards solution

c) $f(x) = 2\cos^{-1}\left(\frac{x+1}{3}\right)$

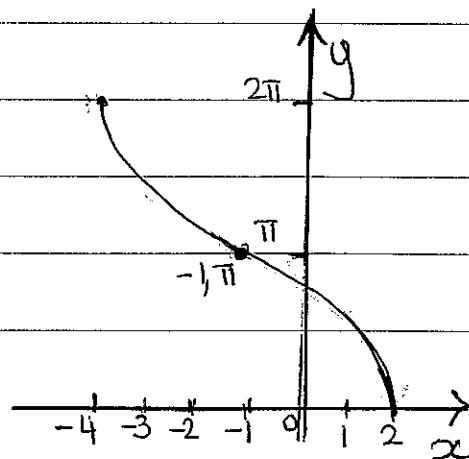
Domain $-1 \leq \frac{x+1}{3} \leq 1$

$$-3 \leq x+1 \leq 3$$

$$-4 \leq x \leq 2$$

Range $0 \leq \cos^{-1}\left(\frac{x+1}{3}\right) \leq \pi$

$$0 \leq 2\cos^{-1}\left(\frac{x+1}{3}\right) \leq 2\pi$$



1 mark states domain and range

1 mark sketches correct curve, correct endpoints

Additional writing space on back page.

Question 6 continues

c ii) $f(x) = 2\cos^{-1}\left(\frac{x+1}{3}\right)$

Inverse function

$$x = 2\cos^{-1}\left(\frac{y+1}{3}\right)$$

$$\frac{x}{2} = \cos^{-1}\left(\frac{y+1}{3}\right)$$

$$\frac{y+1}{3} = \cos \frac{x}{2}$$

$$y+1 = 3\cos \frac{x}{2}$$

$$\therefore f^{-1}(x) = 3\cos \frac{x}{2} - 1$$

1 mark find equation of inverse function

1 mark for correct domain and range

Range of $f(x)$ is domain of $f^{-1}(x)$

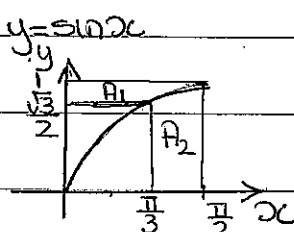
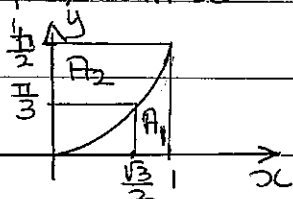
$$0 \leq \frac{x}{2} \leq \pi, \quad 0 \leq x \leq 2\pi$$

Domain of $f(x)$ is range of $f^{-1}(x)$

$$-1 \leq \cos \frac{x}{2} \leq 1, \quad -3 \leq 3\cos \frac{x}{2} \leq 3$$

$$-4 \leq 3\cos \frac{x}{2} \leq 2$$

d) $f(x) = \sin^{-1} x$



$$A_1 = A_{\text{large}} - A_{\text{small}} - A_2$$

$$A_2 = \int_{\pi/3}^{\pi/2} \sin x \, dx$$

$$x=1 \quad f(x) = \sin^{-1} 1 = \frac{\pi}{2}$$

$$A_2 = \left[-\cos x \right]_{\pi/3}^{\pi/2}$$

$$x = \frac{\sqrt{3}}{2} \quad f(x) = \sin^{-1} \frac{\sqrt{3}}{2} = \frac{\pi}{3}$$

$$A_2 = -\left[\cos \frac{\pi}{2} - \cos \frac{\pi}{3} \right]$$

$$A_{\text{large}} = \frac{\pi}{2} \times 1 \text{ sq. unit}$$

$$A_2 = -\left[0 - \frac{1}{2} \right]$$

$$A_{\text{small}} = \frac{\pi}{3} \times \frac{\sqrt{3}}{2}$$

$$A_2 = \frac{1}{2} \text{ square units}$$

$$= \frac{\pi\sqrt{3}}{6} \text{ sq unit}$$

1 mark for finding one of the area of rectangles

1 mark finding A_2

1 mark correct answer with working out

You may ask for an extra Writing Booklet if you need more space.

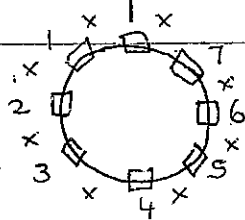
Q6 'd' continues

$$A_2 = \frac{\pi}{2} - \frac{\pi\sqrt{3}}{6} - \frac{1}{2}$$

$$A_2 = \frac{3\pi - \pi\sqrt{3} - 3}{3} \text{ sq units}$$

Q7 a)

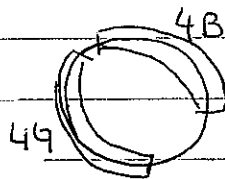
i) Number of arrangements = $7! \times 1$
= 5040



1 mark correct answer with working out.

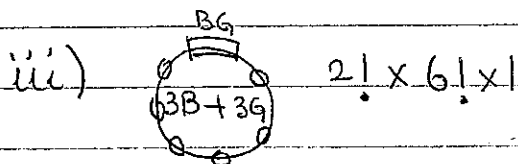
"Bald correct answer does not attract any mark."

ii)



1 mark correct answer with working out

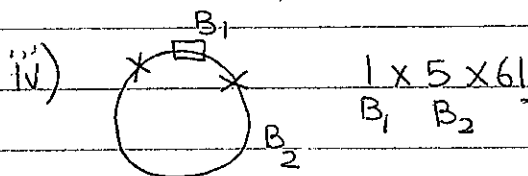
Number of arrangements = $4! \times 4! \times 1$
= 576



No. of arrangements = $2! \times 6! \times 1$
= 1440

2 marks correct solution

1 mark shows either of the 2 distinct parts $2!$ or $6!$



Number of arrangements = $1 \times 5 \times 6!$
= 3600

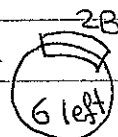
2 marks correct solution

1 mark shows either of the 2 distinct parts.

5 options for B_2

or $6!$ for the rest of the group.

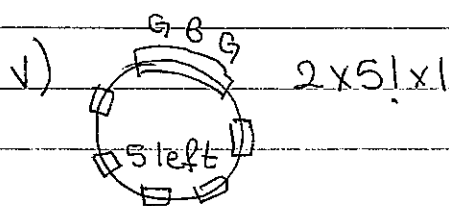
OR If 2 Boys sit together $1 \times 2 \times 6! = 1440$



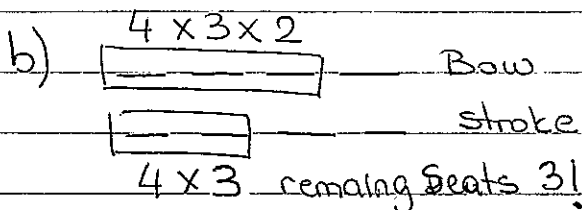
No of arrangements = $5040 - 1440$
= 3600

(Unrestricted - 1440)

Q7 continues



2 marks correct solution
1 mark shows either one of the 2 distinct parts



2 mark correct solution

Number of arrangements = $4 \times 3 \times 2 \times 4 \times 3 \times 3!$
 $= 1728$

1 mark shows either $4P_3$ or $4P_2$ or $3!$

OR $4P_3 \times 4P_2 \times 3! = 1728$

Q8 a) $r = 5\% \text{ pa} = 0.05$

A : Amount each deposit grows.

A_1 : Amount 1st deposit grows in 9 years

A_2 : Amount 2nd deposit grows in 8 years

$A_1 = 500 \times 1.05^9$ 1st deposit at the end of 9 years

$A_2 = 500 \times 1.05^8$ 2nd deposit at the end of 9 years

$A_3 = 500 \times 1.05^7$

$A_4 = 500 \times 1.05^6$

$A_9 = 500 \times 1.05^1$

2 marks: 'Correct' solution

$A_1 + \dots + A_9 = 500 (1.05 + 1.05^2 + \dots + 1.05^8 + 1.05^9)$

Geometric series $a = 1.05$
 $r = 1.05$

1 mark correct reasoning for excluding A_3 and A_4 or adding $\$1000 \times 1.05^6$

$\Sigma A = 500 \frac{1.05(1.05^9 - 1)}{1.05 - 1}$
 $= \$5788.94$

Q8 continues

But she missed A_3 and invested \$1000

in 4th year. Therefore the amount at the end of 9 years

$$A = 5788.94 - 500 \times 1.05^7 - 500 \times 1.05^6 + 1000 \times 1.05^6$$

$$A = \$5755.44$$

b) i) $r = 10\% = 0.1$

Amount at the end of 1st year:

$$A_1 = 20000(1+0.1) - 2500$$

$$A_2 = (20000 \times 1.1 - 2500)1.1 - 2500$$

$$= 20000 \times 1.1^2 - 2500 \times 1.1 - 2500$$

$$A_2 = \$[20000 \times 1.1^2 - 2500(1.1 + 1)]$$

1 mark correctly shows A_2

ii) Using the result from part (i)

$$A_n = 20000 \times 1.1^n - 2500(1 + 1.1 + \dots + 1.1^{n-1})$$

geometric progression

$a=1$ $r=1.1$, n terms

1 mark for writing A_n using part (i)

$$A_n = 20000 \times 1.1^n - 2500 \frac{1(1.1^n - 1)}{1.1 - 1}$$

$$A_n = 20000 \times 1.1^n - 25000(1.1^n - 1)$$

A_n should be greater than \$2500 so that she can go on holiday.

$$2500 < 20000 \times 1.1^n - 25000(1.1^n - 1)$$

$$1 < 8 \times 1.1^n - 10 \times 1.1^n + 10$$

$$2 \times 1.1^n < 9$$

1 mark for reducing the equation into

$$1.1^n < 9$$

1 mark for correct solution

$$1.1^n < 4.5$$

$$\text{If } 1.1^n = 4.5$$

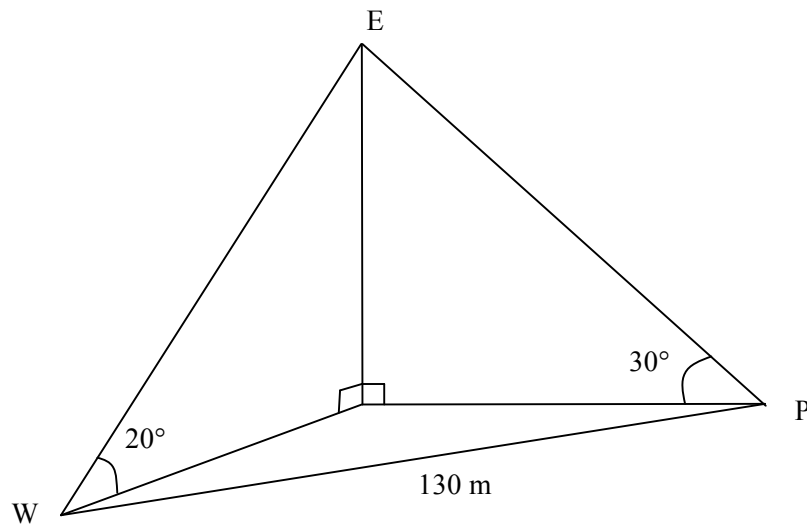
$$n \ln 1.1 = \ln 4.5$$

$$n = \frac{\ln 4.5}{\ln 1.1} \quad n = 15 \quad A_{15} = \$4113.76$$

$$n = 15.8 \quad n = 16 \quad A_{16} = \$2025$$

$$\therefore n = 16 \text{ years}$$

- (d) A bird watcher can see an eagle high up in a tall tree due north of him. The angle of elevation to the eagle is 20° . From a different site, 130 metres along a straight, horizontal road, a photographer can see the same eagle, due west of her position, at an angle of elevation of 30° .

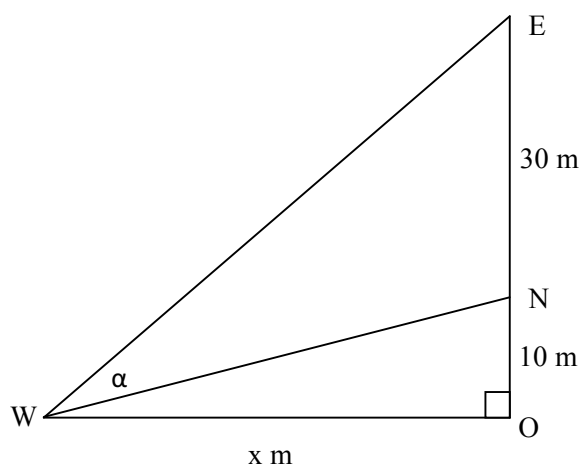


- i) Show that the eagle is 40 metres above the ground.

3

Question 6d continues

The bird watcher moves to a new position and can now see an eagle's nest on a branch 10 metres above the level of his eyes and directly below the eagle.



When the bird watcher is standing x metres from the base of the tree, the size of angle $\angle EWN$ is given by

$$\alpha = \tan^{-1} \frac{40}{x} - \tan^{-1} \frac{10}{x}$$

(Do NOT prove this.)

ii) Show that

2

$$\frac{d\alpha}{dx} = \frac{-40}{x^2 + 1600} + \frac{10}{x^2 + 100}$$

(iii) Hence, find how far from the base of the tree the bird watcher must stand in order for the angle α to be greatest.

2

End of Question 6

- (a) Use mathematical induction to show that $5^n > 3^n + 4^n$ for all positive integers $n \geq 3$. **3**
- (b) Find the number of ways that 12 different beads can be arranged on a bracelet. **1**
- (c) (i) How many eight-letter arrangements of the letters in the word GIGGLING are possible? **1**
- (ii) What is the probability that the letters L and N will be next to each other? **2**
- (d) In how many ways can 4 boys and 3 girls be arranged in a row if the selection is made from 6 boys and 5 girls? **2**
- (e) A supermarket has 5 self-serve checkouts. Two of the checkouts only accept cash while the other three accept cash or credit card. Three shoppers use the checkouts to make their purchases.
- (i) One shopper must use a credit card while the other two can use cash or a credit card. Find the number of ways that the three shoppers can make their purchases so that they all use different checkouts. **1**
- (ii) On another day, all three shoppers have cash. Find the number of ways in which the three shoppers can make their purchases if at least one uses a cash only checkout and each checkout can be used more than once. **2**

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

STUDENT NUMBER

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Section I

5 marks
Attempt Questions 1–5
Allow about 8 minutes for this section

Select the alternative A, B, C or D that best answers the question and indicate your choice with a cross (X) in the appropriate space on the grid below.

	A	B	C	D
1				
2				
3				
4				
5				